INCH-POUND

A-A-58000
16 MAR 1992

COMMERCIAL ITEM DESCRIPTION

TANK, STORAGE, LIQUID ARGON-NITROGEN-OXYGEN, Cryogenic, Horizontal, 3000 and 6000 Gallon

The General Services Administration has authorized the use of this Commercial Item Description.

1. SCOPE

1.1 Scope. This Commercial Item Description (CID), describes a 3000 and 6000 gallon horizontal cryogenic storage tank capable of storing and delivering liquid oxygen or liquid nitrogen.

1.2 Classification. The storage tanks shall be of the following types and sizes. The type and size of storage tank to be furnished shall be as specified (see 6.1).

Type I - Liquid Oxygen

Type II - Liquid Nitrogen

Size A - 3000 Gallon

Size B - 6000 Gallon

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: SA-ALC/SFRT Kelly AFB TX 78241-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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FSC 3655

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2. APPLICABLE DOCUMENTS

2.1 Government Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this description to the extent specified herein.

Specifications

Military

MIL-H-27301 - Hose Assembly, Metal, Cryogenic Liquid, Transfer

Standards

Military

MIL-STD-130 - Identification, Marking of U.S. Military Property

Federal

FED-STD-376 - Preferred Metric Units For General Use By The Federal Government
FED-STD-595 - Colors

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, ATTN: NPODS, 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

Drawings

Ordnance Corps

8987831	Assembly, LOX Coupling, Female Half, All Sizes
8987840	Retaining Ring, 1-Inch
8987856	Cone, 1-Inch
8987857	Nut, 1-Inch
8987870	Plug, 1-Inch
8987877	Cone, 1-Inch
8987878	Nut, 1 Inch
8987889	Assembly, Liquid Nitrogen Coupling, Female Half, All Sizes
8987891	Plug, 1-Inch
8991172	Assembly, Liquid Nitrogen Coupling, Female Half, 1-Inch
8991174	Assembly, LOX Coupling, Female Half, 1-Inch

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Air Force

8221044-01	Decal, Liquid Oxygen
8221045-01	Decal, Liquid Nitrogen
8321391	Filter Assembly, 1-Inch
8518507	Filter, Assembly, 2-Inch
8752448-01	Nameplate

(Copies of Drawings required by suppliers in connection with specific acquisition functions shall be obtained from the contracting activity, or as directed by the contracting officer.)

Federal Acquisition Regulation (FAR)

(Unless otherwise indicated, copies of the Federal Acquisition Regulation (FAR) are available by contacting, Superintendent of Documents, Congressional Sales Office, U.S. Government Printing Office, Washington, DC 20402)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society of Mechanical Engineers

ASME Boiler and Pressure Vessel Code, Section VIII, Div 1

(Copies may be purchased from American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017)

Phamphlet

G-4.1	Cleaning Equipment for Oxygen Service
V-6	Standard Cryogenic Liquid Transfer Connections

(Unless otherwise indicated, copies shall be obtained from, Compressed Gas Association Inc., 1235 Jefferson Davis Highway, Arlington VA 22202)

2.3 Order of Precedence. In the event of a conflict between the text of this CID and the references cited herein, the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. SALIENT CHARACTERISTICS

3.1 Design. The storage tanks shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The storage tanks shall meet the requirements specified herein and shall conform to current commercial industrial practice.

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3.2 Capacity. The storage tanks shall have a net volume of 3000 gallons (+) or (-) 2 percent or 6000 gallons (+) or (-) 2 percent (see 6.1). The tank shall have a gross water volume rating of not less than 102 percent of the rated net volume.

3.3 Dimensions. The envelope dimensions of the tank shall not exceed 9 feet in width and 10 feet in height. The length for the 3000 and 6000 (see 6.1) gallon tank, including the plumbing, shall not exceed 20 feet and 32 feet, respectively.

3.4 Capabilities. The tank shall be capable of storing and delivering liquid oxygen or liquid nitrogen (see 6.1). The tank shall be designed and fabricated per ASME, Boiler and Pressure Vessel Code, Section VIII, Div 1 for a maximum allowable working pressure of not less than 55 psig. The tank shall incorporate an integral pressure building unit with adequate heat absorption characteristics for the respective size tank.

3.5 Plumbing Requirements. The tank shall have plumbing lines and associated valves to include, as a minimum, top and bottom fill, liquid withdrawal, vent, full trycock, pressure building unit and instrumentation.

3.5.1 Top and Bottom Fill Line. The top and bottom fill line shall incorporate a 2-inch filter, per AF Drawing 8518507, upstream of the fill line valve. The filter shall be incorporated into the plumbing system to facilitate removal and replacement. The top and bottom fill line shall terminate in a Standard Connection, OX-150 or NI-150 (see 6.1), IAW CGA Pamphlet V-6, Standard Cryogenic Liquid Transfer Connections. A fill line safety valve, a fill line drain valve and a fill line check valve shall also be required.

3.5.2 Liquid Withdrawal Line. The liquid withdrawal line shall incorporate a 1-inch filter, per AF Drawing 8321391, downstream of the liquid withdrawal line valve. This filter shall also be incorporated into the plumbing system to facilitate removal and replacement. The liquid withdrawal line shall terminate in a 1-inch male NPT nipple to facilitate the attachment of the hose specified in 3.8. A withdrawal line safety valve and a withdrawal line drain valve shall also be required.

3.6 Protection Devices. Pressure relief devices shall be provided for protection of the inner tank and the outer shell. The inner tank shall have, as minimum, a primary and secondary relief device sized and set per ASME Code, section VIII, Division 1 to allow for a normal operating pressure of 50 psig.

3.7 Instrumentation. All instruments and isolation valves necessary for effective and safe operation of the tank shall be provided. Specific marking requirements for the pressure gage and liquid level gage are as follows:

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3.7.1 Liquid Level Gage. The liquid level gage shall be scaled to indicate gallons of liquid. The scale shall be of the dual reading type, capable of reading liquid oxygen and liquid nitrogen. (This dual reading gage is required irregardless of the type tank specified in 6.1.) The outer band shall indicate and be marked for oxygen. The inner band shall indicate and be marked for nitrogen. Both bands, inner and outer, shall be green from 0 gallons to the net volume of the size tank specified (see 6.1) and shall be red from the net volume to the gross volume of the size tank specified (see 6.1). The scale range shall be from 0 to the gross volume of oxygen for the size tank specified (see 6.1). The scale shall have graduations every 200 gallons with 50 gallon increments. The gage face shall also be marked in red lettering, "USE NO OIL".

3.7.2 Pressure Gage. The pressure gage provided shall have a range from 0 to 100 psi. The gage shall have a green color band from 0 to 50 psi and a red color band from 50 to 100 psi. The scale shall have graduations every 10 psi with 1 psi increments. The gage face shall also be marked in red lettering, "USE NO OIL".

3.8 Liquid Withdrawal Hose and Storage Provisions. A liquid withdrawal hose conforming to MIL-H-27301 (see 6.1) and terminating in a 1-inch liquid oxygen or liquid nitrogen female ordnance coupling assembly (see 2.1 Drawings, Ordnance Corps and see 6.1) on one end shall be provided with the tank. Storage provisions for the hose assembly shall also be provided on the tank for transportation purposes.

3.9 Color and Markings. The final finish paint coating color shall be gloss white number 17875 per FED-STD-595. All valves and gages shall be identified by securely attached nameplates to reflect the nomenclature or symbol utilized in the operating instructions. The tank shall be marked for identification in accordance with MIL-STD-130. The tank nameplate 8752448-1 (see 6.1) shall be securely attached to the tank in a readily visible location. Each tank shall require decals, 8221044-01 for liquid oxygen and 8221045-01 for liquid nitrogen (see 6.1), applied one on each side of the tank at the middle above the horizontal center line of the tank, and one centered on each end of the tank.

3.10 Cleanliness. All surfaces, parts, fittings, etc. that will be in contact with oxygen or nitrogen shall be thoroughly cleaned as specified in CGA Pamphlet G-4.1 to remove all metal shavings, oil, grease, and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor certification. The contractor shall certify, and maintain objective quality evidence, that the product offered meets this commercial item description, and that the

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product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market Acceptability. The item offered must have been sold to the government or commercial market. The item must have been on the market for at least 5 years to determine reliability and performance of the item. A modified commercial, off-the-shelf item is required to minimize design and engineering risk, to assure that an established end product is routinely supported by spare and repair parts and to preclude untested or experimental units.

4.3 Responsibility of inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.4 Testing. Documentation of testing (see 4.5, 4.6, 4.7, 4.8) when required shall be as specified by the contracting officer.

4.5 Examination. Each tank shall be visually examined to determine conformance with all requirements of this description.

4.6 Pressure test. Each component and circuit of the tank that operates under positive pressure in normal service shall be pressure tested in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code. Any distortion, damage, or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

4.7 Vacuum retention test. The annular insulation space absolute pressure shall be determined and verified to comply with the limits established by the manufacturer.

4.8 Cleanliness test. The cleanliness tests apecified in CGA Pamphlet G-4.1 per para 11.2.1, 11.2.2, 11.2.5 and 11.2.6 shall be performed utilizing the pass/fail criteria specified by the applicable test.

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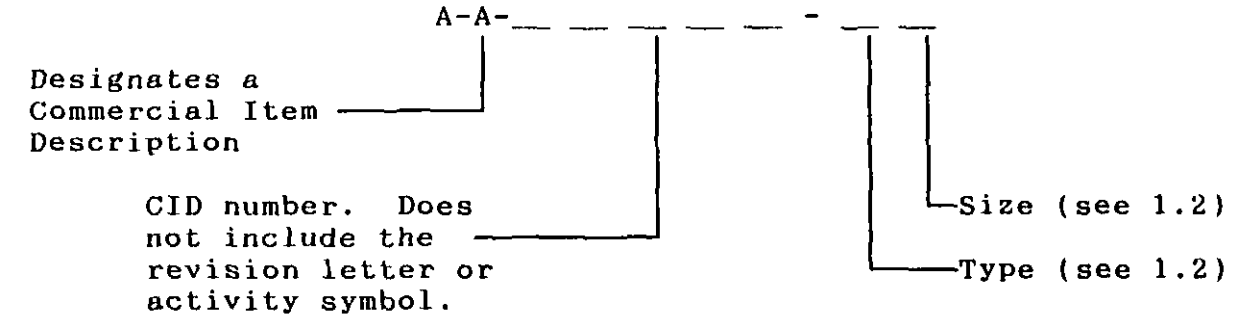
4.9 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4.10 Regulatory Requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5. PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

5.1 Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

5.2 CID based part identification numbers. The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



6. NOTES

6.1 Acquisition. Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description.

b. Type, size, and quantity required (see 1.2).

c. Provisions for government furnished material.

d. Packaging requirements.

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6.2 Suppliers.

Minnesota Valley Engineering Inc.
407 Seventh St. N.W.
New Prague, MN 56071

Process Engineering
Plaistow, New Hampshire 03865-0467

6.3 CUSTODIAN	PREPARING ACTIVITY
Air Force-99	Air Force-68

PROJECT 3655-0121

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